Development and Characterization of Rubidium Atomic Oven Under Varying Pressure

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Abstract:

The proposed project aims to build and characterize a rubidium (Rb) atomic oven to supply a high density of dopants to a large, single crystal of argon (Ar) under varying oven pressures. These doped crystals could be used to make measurements with high statistical sensitivity of the electric dipole moment (EDM) of fundamental particles to search for physics beyond the standard model. The Odom group hopes to define and optimize the growth of these crystals in many different conditions while maintaining a dopant density of around 10^{13} atoms per cubic millimeter. Therefore, the oven will need to be designed and characterized to allow for a consistent supply of dopants while the crystal growth chamber varies in pressure and temperature. By employing a microcapillary array, I aim to collimate the beam of Rb leaving the oven to prevent beam divergence and increase atomic flux (atoms passing a unit area every second), allowing for higher dopant density and offsetting the effects of changes in pressure. The three-part oven design will allow for easy modifications and have individual temperature control on each part to fine-tune atomic flux and beam divergence. The oven and test growth chamber will also allow for saturated absorption spectroscopy (SAS) to precisely measure atomic flux and beam divergence for oven characterization. This project will be an important part of ongoing research in the Odom group in the development of new ways to make more precise measurements to advance knowledge of fundamental physics.

Where Work Will Be Done and New Knowledge Produced:

The research will be conducted in the Odom Group laboratory located in Northwestern's Center for Fundamental Physics (CFP) located in Mudd Library, benefiting from the highly collaborative environment of the CFP. Atomic ovens are common, but our oven must deal with circumstances not nearly as common or previously characterized. Since the first step of our main experiment is finding the best way to grow large, doped crystals, we will vary multiple parameters, the most impactful of which for the oven will be the pressure of our chamber. During this process, we may use pressures ranging from 10^{-6} torr to one torr. However, most Rb ovens are designed to operate at one pressure, usually between 10^{-10} torr to 10^{-5} torr [2][3]. Operating at higher pressures is not necessarily a problem, though, since this just requires us to run the oven at a hotter temperature. This increases the vapor pressure/amount of Rb in the oven to maintain the ratio of Ar to Rb associated with our desired dopant density. People have done this in similar crystal growth experiments to operate ovens at pressures above one torr [4]; however, they still maintained a constant pressure, so the effect of actively varying the pressure in an atomic oven is still poorly characterized. This project will characterize the effects of varying pressure on an atomic oven and provide information helpful to future experiments in the Odom lab and other similar experiments.

Work Description:

The proposed work focuses on the development and characterization of a rubidium atomic oven to achieve high, consistent dopant densities in experiments to optimize the growth of large, doped, single argon crystals. These crystals could be used to make more precise measurements of the electric dipole moment (EDM) of fundamental particles to look for physics beyond the standard model. Finding EDM values not predicted by the standard model could provide insight into some of the biggest mysteries in physics, such as matter-antimatter asymmetry. To make these types of measurements, we hope to reach dopant densities–supplied by our oven–of 10^{13} atoms per cubic millimeter in our crystals. Additionally, when optimizing crystal growth, we will grow and dope these crystals at multiple pressures. Therefore, I will need to characterize how changing the pressure affects dopant density and then determine how to maintain our desired dopant density regardless of the pressure at which we operate.

My project will begin with building an atomic oven that can supply our dopant density and have refined and varying control over temperature to affect the amount of dopants in our oven. The proposed atomic oven design consists of three main components: a sidearm to hold the Rb sample, a central chamber with windows to allow for saturated absorption spectroscopy (SAS), and a nozzle featuring a microcapillary array to collimate the Rb beam. Each section will have its own heating element to allow for more nuanced control over oven temperature. Heating the sidearm mainly affects how much Rb we have in our oven at any time, while heating the nozzle largely affects the atomic flux and beam divergence (angle our beam deviates from a straight path) of the Rb leaving our oven. We use microcapillaries because they have been shown to create collimated atom beams with divergence angles of around only 1.2 degrees and have also been shown to produce a high atomic flux of around $7x10^{11}$ atoms/sec per microcapillary [1][2].

To start characterizing the oven, I will connect it to a test chamber with additional sets of windows to allow for SAS at the nozzle of the oven and the growth plate location. This, along with the windows in the oven, will allow us to see how much Rb is at each point in our chamber. SAS has also been used to characterize atomic beams by finding atomic flux and defining beam divergence [5]. My first task will be to make the Rb beam as collimated as possible by varying the different temperatures of each part of the oven; it is common to have the nozzle of ovens hotter than the central chamber of the oven [1][2][3], so I will start with this. I aim to match the previously mentioned beam divergence of 1.2 degrees at the growth plate and atomic flux of order 10^{14} leaving the nozzle at a low chamber pressure (around 10^{-7} torr). I will also use this phase of my project to ensure my design can reach these criteria; if divergence is too large and flux too low, I can modify the microcapillary array by using thinner and longer capillaries or increasing the temperature of the nozzle, both methods should decrease divergence and increase flux. These changes will be easy due to the modular three-part design. Once I meet these specifications, I will maintain temperatures and increase pressure while continuing to measure beam divergence and atomic flux. This data will give me a better understanding of how sensitive my oven is to chamber pressure. Next, I will maintain pressure while varying oven temperature and continue to measure flux and divergence. With these two data sets, I can see how flux and divergence are offset from our target at a certain pressure, and then I can look at temperature data to see how much I need to change temperature to undo this offset. This method is unlikely to

work exactly, but it will provide a starting point at which, if still off of the desired divergence and flux, I can change temperature beyond what the temperature data predicts. Doing this several times will provide insight into how ovens operate at varying pressures and allow me to identify how to maintain our desired dopant density for all the pressures we work with.

Value of Award:

This award would be immensely valuable to me, both personally and professionally. The financial support provided by the award would allow me to put my entire focus and dedication toward the project without the distraction and worries of personal financial constraints. This project is the first time I will be the main person in charge of a project and making important decisions that will impact my entire lab, which will give me a lot of experience and decision-making skills needed to pursue a future career in research. Additionally, the work I will do under this award could be used in part of a senior thesis. Overall, this award will allow me to undertake a project that will give me experience that will be crucial in my continued pursuit of a career in physics and research.

Personal Resume:

Last spring and summer, I worked in Professor Kovachy's lab designing a vapor cell, which, like an oven, reaches high temperatures to get atoms, like Rb, into a gaseous phase. During this project, I spent several hours designing the cell in Solidworks, machining prototype parts, and testing different heating elements. All of these skills will be helpful in my oven project. During that summer I also received funding from the Weinberg Undergraduate Research Grant. Additionally, the Ph.D. student I worked with during that project has experience with ovens and expressed her willingness to help me with any questions that arise during this new project. I have been working in Professor Odom's lab since the fall quarter, helping design the main crystal growth chamber and the oven. I attend weekly meetings, both with the entire lab and with a Ph.D. student, allowing me to get an in-depth understanding of the overall goals of the lab and my project. Additionally, with the completion of Physics 339-1/339-2 (Quantum Mechanics) by summer, I will have a much better general understanding of the physics our lab will work with, and with the completion of Physics 332-0 (Statistical Mechanics), I will have a much better understanding on how to do calculations to understand how Rb and Ar behave in both our oven and crystal growth chamber.

References

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